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Keywords

Domain knowledge graph, Semantic retrieval, Large Language Model, Multi-objective screening, Optoelectronic materials

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OptoMatKG: A Traceable Knowledge Graph Framework with Semantic Retrieval for Optoelectronic Materials Science

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Abstract

Extracting and correlating knowledge from heterogeneous scientific publications poses a fundamental data engineering challenge for data-driven materials discovery. Traditional keyword search lacks semantic depth, while generic large language models (LLMs) are prone to hallucinated predictions. To address these limitations, we present OptoMatKG, a literature-scale data engineering system whose core contributions are threefold: (1) a domain-specific knowledge graph construction pipeline based on a nine-category full-workflow ontology; the resulting graph comprises 2.62 million entities and 3.38 million typed relations; (2) a cascaded multi-stage retrieval pipeline that synergistically integrates LLM-based query normalization, semantic vector indexing and Materials Project pre-filtering to enhance metric-aware candidate ranking; and (3) an evidence-constrained auditing module that forces every output to anchor to DOI-supported reasoning chains, thereby enforcing end-to-end hallucination prevention. Using optoelectronic inverse design as a validation testbed, experiments verify that our entity extractor reaches a micro-averaged F1 of 0.971. Comprehensive tests over 2000 real-world queries further demonstrate that our system surpasses keyword-based and single-stage baselines in Recall@K and metric coverage. The system accurately quantifies multi-objective performance trade-offs while ensuring fully traceable inference. Although validated on photodetectors, the proposed solution provides a generalizable, pluggable engineering framework readily extendable to other functional material discovery pipelines.

1 Introduction

The rapid proliferation of heterogeneous scientific literature has created a fundamental data engineering challenge: how to systematically extract, structure, and retrieve cross-referenced knowledge from millions of published studies. In the domain of optoelectronic materials discovery, this challenge is particularly acute. [1–4] Traditional research workflows rely on manual operation of computational simulation packages such as VASP[5], Quantum-ESPRESSO[6], and Atomic Simulation Environment (ASE)[7], alongside static crystal databases including the Materials Project[8] and Open Quantum Materials Database (OQMD)[9], all supplemented by time-consuming manual reading of experimental papers. These workflows require extensive expertise to correlate material compositions, fabrication processes, microstructural features, characterization techniques, and device performance metrics—a labor-intensive process that suffers from high personal expertise dependence, low replicability, and frequent omission of cross-dimensional experimental evidence.[10–12]

Existing computational approaches to accelerating this process fall into three categories: (1) high-throughput density functional theory (DFT) calculations, which are computationally expensive and often fail to capture synthesis-related experimental conditions; (2) machine learning surrogate models trained on static crystal databases, which generalize poorly to novel composition spaces and provide limited traceability to original experimental

evidence; and (3) keyword-based literature retrieval systems, which fail to capture domain-specific semantic relationships and cannot support multi-objective trade-off queries. None of these paradigms adequately addresses the dual requirements of optoelectronic device design: retrieving evidence-supported candidate materials while providing auditable reasoning chains back to primary literature, with explicit marking of unreported metrics.

To overcome these limitations, we propose OptoMatKG, a literature-grounded knowledge graph and semantic retrieval framework for traceable, evidence-constrained multi-objective screening of optoelectronic materials. The core contributions of this work are threefold:

(1) We construct a full-process optoelectronic ontology and a DOI-provenanced knowledge graph spanning experimental synthesis, fabrication processes, characterization methods, measurement data, and device performance metrics. This graph enables auditable cross-link reasoning across the entire material research pipeline.

(2) We develop a cascaded multi-stage retrieval pipeline that integrates LLM-driven query normalization, semantic vector indexing, Materials Project pre-filtering, and metric-aware graph supplementation. This pipeline improves the ranking of candidates under complex multi-objective design constraints by adhering to evidence-supported metrics only, excluding missing values from constraint satisfaction.

(3) We design an evidence-constrained auditing module that links retrieved numerical values to verified graph records and primary literature DOIs. For any unreported metric, the module outputs a missing indicator (i.e., missing values are not used for multi-objective constraint satisfaction), while extracting interpretable reasoning paths to mitigate hallucinatory quantitative claims.

2 Related Work

The related studies are divided into the following three main lines.

Domain-specific material knowledge graphs. Static crystal databases mainly store crystallographic and thermodynamic parameters, while lacking unified structured descriptions of synthesis routes, fabrication processes, characterization protocols and device-level optoelectronic performances[13]. Literature-mined knowledge graphs such as Perovskite KG[14] and MatKG[15] generally adopt shallow ontology schemas or simple co-occurrence-based relational links. They cannot construct complete end-to-end chains covering the full experimental workflow, and few of them retain permanent DOI provenance for quantitative performance verification[16]. Consequently, existing material graphs fail to support traceable, fully evidence-backed reasoning spanning material composition, fabrication processing and device performance[17], which is essential for optoelectronic multi-objective screening.

Domain-adapted semantic retrieval strategies. Domain-pre-trained text encoders including SciBERT[18], MatSciBERT[19] and MatBERT[20] outperform traditional keyword matching, yet mainstream retrieval pipelines only index abstracts and superficial text mentions. They return unstructured text snippets instead of structured metric-aligned graph evidence.[21] Single-stage ranking merely optimizes overall text similarity, which severely impairs retrieval accuracy for narrow-band optoelectronic metrics decisive to multi-objective device screening.[22] Though auxiliary clustering or re-ranking modules are occasionally introduced, few frameworks adopt cascaded multi-stage

filtering that fuses database-derived candidates and graph recall results, inevitably losing optimal materials targeting specific performance metrics.[23]

LLM-based literature mining and hallucination suppression. Retrieval-augmented generation (RAG) has become a mainstream paradigm for literature analysis.[24–26] However, most existing systems lack standardized optoelectronic ontologies for cross-system performance comparison, hindering quantitative evaluation of performance gaps between target requirements and candidate materials.[27, 28] Besides, retrieval and ranking modules are rarely customized for multi-objective optoelectronic demands. Without complete evidence tracing mechanisms, LLMs tend to generate unsubstantiated numerical predictions lacking valid literature citations.[29] Current anti-hallucination strategies mostly rely on isolated post-hoc validation, rather than an end-to-end audit workflow that binds all predicted numerical results to graph records and original DOI sources.[30,31] Moreover, graph-augmented LLM frameworks seldom explicitly label unreported performance metrics or generate human-readable evidence-supported reasoning chains required for optoelectronic inverse design screening.[32]

In summary, existing approaches exhibit three primary deficiencies: (1) knowledge graphs lack full-workflow coverage and DOI-provenanced traceability; (2) retrieval pipelines fail to support metric-aware multi-stage filtering; and (3) LLM-based systems lack evidence-constrained auditing to prevent hallucinated predictions.

3 Methodology

Figure 1 presents the optoelectronic knowledge graph, which stores structured KG records containing domain entities and typed relational links. User design queries and graph records are fused via joint feature fusion, then fed into a domain pre-trained MatBERT model with shared encoder weights. The model generates normalized query vectors for user demands and record vectors for literature materials. All record vectors pass Materials Project pre-filtering and are stored in two offline databases: vector index for fine-grained similarity matching and semantic clustering index for rapid coarse pre-filtering during retrieval. In the middle-right module, cosine similarity semantic retrieval obtains Top-K candidate materials from the vector index assisted by clustering pre-filtering.The retrieved candidates are imported into the evidence-constrained audit module, which sequentially implements DOI provenance verification, quantitative performance gap analysis, labeling of unreported performance metrics, and genetic algorithm (GA) multi-objective optimization, finally outputting interpretable, hallucination-free optoelectronic material candidates for photodetector design.

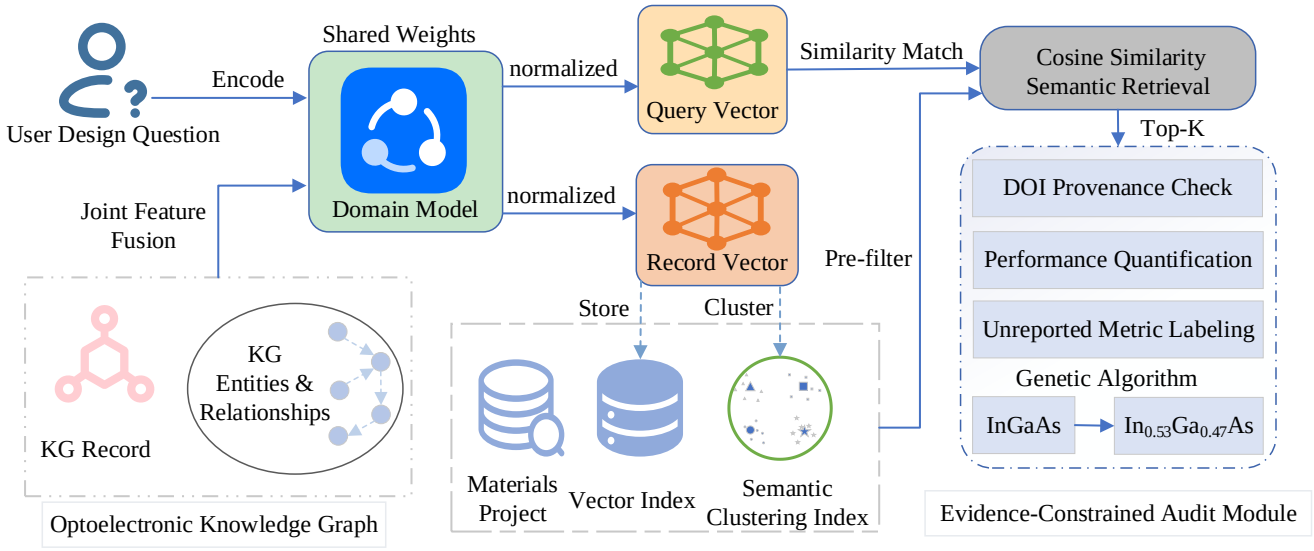


Figure 1. Framework of OptoMatKG retrieval: field-aware query normalization, vector Top-K search, evidence-constrained DOI filtering, and semantic grouping for traceable candidate recommendation.

3.1 Knowledge Graph Construction

We compiled a raw corpus containing 75,979 peer-reviewed optoelectronic papers. The dataset covers photodetectors, light-emitting diodes, solar cells and other mainstream optoelectronic devices. The complete pipeline for knowledge graph construction and data import is visualized in Figure 2.

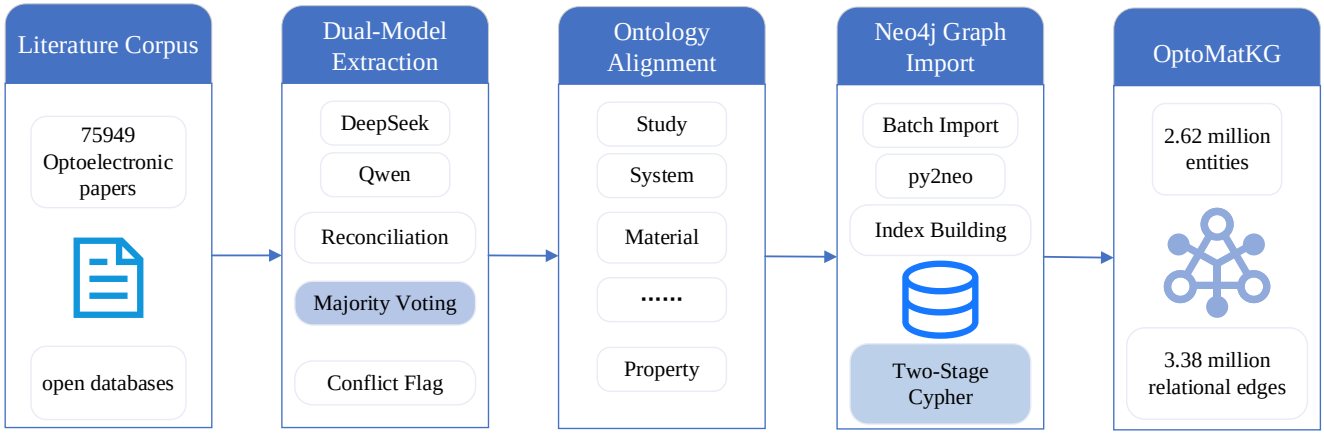


Figure 2. Schematic workflow of OptoMatKG knowledge graph construction and data import. Optoelectronic papers are processed by a dual-model LLM extraction module, reconciled via majority voting with conflict marking, and aligned to a ten-category ontology.

To standardize extracted information, we establish a dedicated ontology with 9 mutually exclusive core entity types and 10 standardized bidirectional semantic predicates. The logical architecture of this ontology, namely the graph schema, is illustrated in Figure 3.

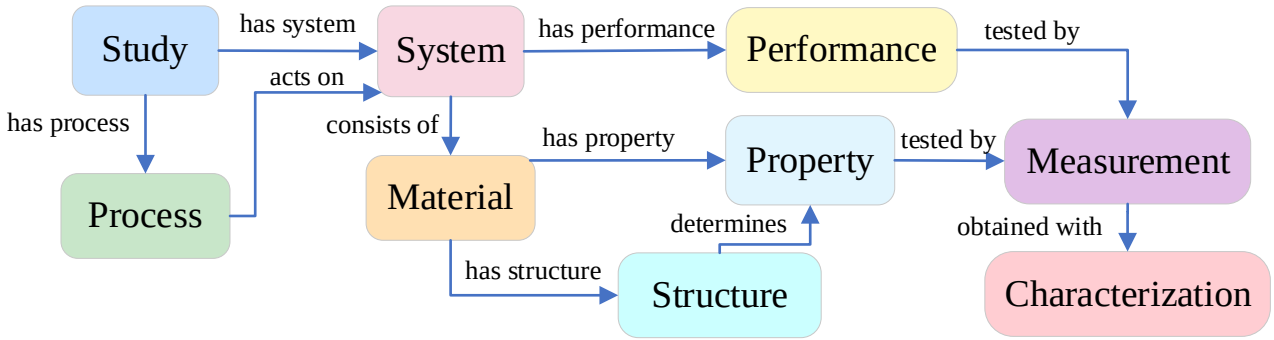


Figure 3. Knowledge graph schema for optoelectronic materials literature. Each node class denotes a distinct materials-science concept; each directed edge denotes a typed semantic relation.

3.2 Semantic Indexing with MatBERT

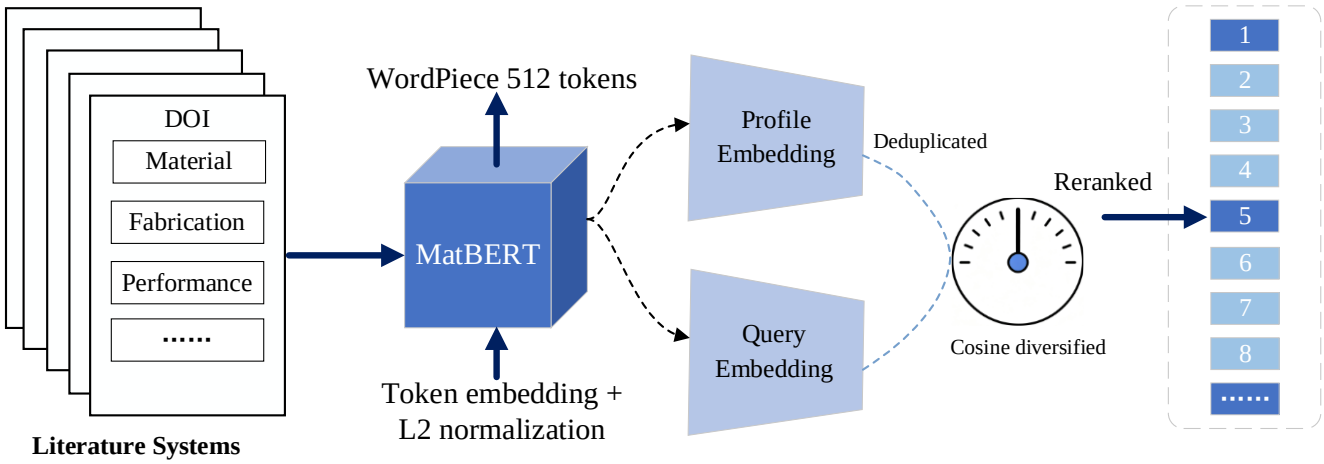


Figure 4. MatBERT-based semantic indexing and candidate reranking in OptoMatKG.

Literature-derived system profiles (DOI, material, fabrication, and performance fields) are encoded by MatBERT using WordPiece tokenization (maximum length 512). Token embeddings are pooled via [CLS] pooling into a fixed-length system embedding, followed by L2 normalization. User queries are projected into the same semantic space, and cosine similarity is used to retrieve Top-K candidate systems. Retrieved candidates are then refined by semantic reranking, including DOI deduplication and diversification via maximal marginal relevance (MMR) to reduce redundancy, yielding semantically aligned candidate sets for downstream evidence-constrained graph auditing.

Optional clustering for latency optimization. To reduce online retrieval latency, we optionally apply K-means clustering over system embeddings. During inference, cosine similarity is computed only against nearest cluster centroids to accelerate nearest-neighbor search.

3.3 Multi-Stage Retrieval and Evidence-Constrained Audit

We propose a cascaded multi-stage semantic retrieval pipeline to map natural-language material design requests to DOI-traceable candidates, equipped with an evidence-limited audit module to avoid LLM hallucinations. The

pipeline integrates metric-aware graph supplementation, Materials Project pre-filtering, Neo4j keyword supplementation and candidate diversification to balance multi-objective retrieval recall and diversity. Frontend caching is used to lower inference cost, and ablation tests are conducted with identical core retrieval configurations (audit module disabled for all baselines). Detailed hyperparameters and module logic are listed in Supporting Information Tables S1–S2.

Retrieved top-10 candidates are further refined by a four-step audit workflow: metric normalization and gap classification, domain-aware candidate prioritization, full attribute deep-fetching from Neo4j, and evidence-constrained LLM summary generation. To eliminate unfounded predictions, we embed multiple anti-hallucination strategies: unmeasured metrics are not inferred, fake values and invalid DOIs are discarded.

3.4 Genetic Algorithm-guided Multi-objective Candidate Refinement

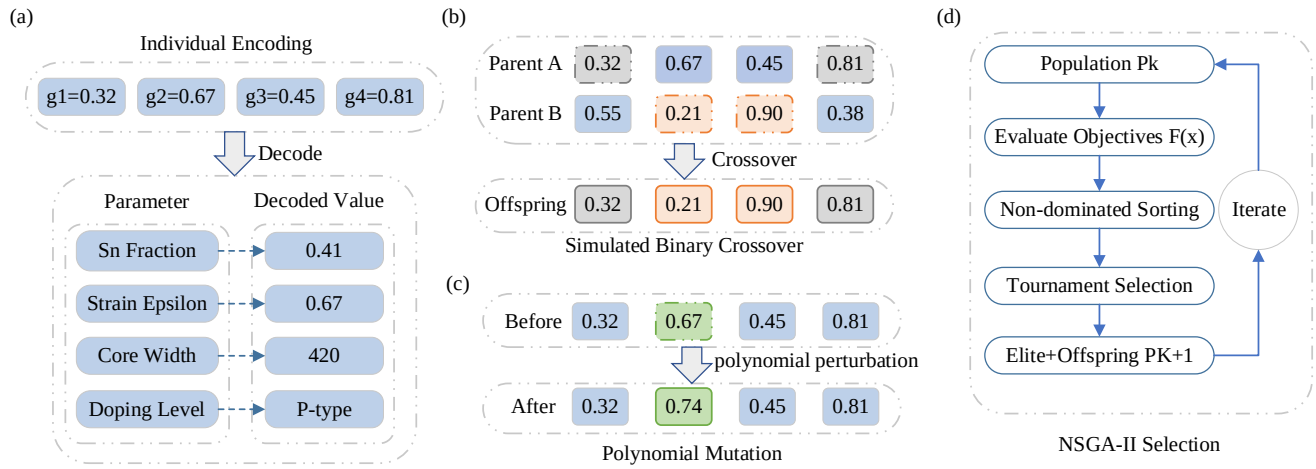


Figure 5. Schematic of genetic algorithm (GA) operators in OptoMatKG: individual encoding, crossover, mutation, and multi-objective selection or update. Candidate designs are evolved for 30 generations to obtain a Pareto front of non-dominated solutions.

To improve multi-objective trade-offs after semantic retrieval, we optionally run a continuous-variable GA. Each individual is encoded as a normalized gene vector over a constrained design space (composition, strain, and device-geometry variables), then decoded into physically interpretable parameters. The design-space bounds are informed by validated material proposals and physics checks rather than direct fitting to DOI-audited numeric ranges.

For material-level search, the objective vector F is minimized using bandgap mismatch and optional losses. In photodetector mode, response time, dark current, and detectivity are evaluated via surrogate proxy models (order-of-magnitude device estimates), not by directly optimizing DOI-audited experimental values. Evolution uses non-dominated sorting genetic algorithm II (NSGA-II)[33], with tournament selection, simulated binary crossover (SBX)[34], polynomial mutation (PM; distribution index $\eta = 15$)[35], and elitist update from P_k to P_{k+1} over 30 generations (population size 40). The operators are schematized in Figure 5. Importantly, the GA output does not fabricate literature evidence; it proposes proxy-level design hypotheses that complement, rather than replace, DOI-audited screening under conflicting objective.

4 Experiments and Results

We evaluate OptoMatKG along six complementary dimensions: (1) experimental protocol covering datasets, baselines and evaluation metrics; (2) dual-model entity extraction performance; (3) scale and schema coverage of the constructed knowledge graph; (4) retrieval ablation experiments based on 2000 design queries; (5) fine-grained SiC case analysis and a 1550 nm photodetector screening case, which distinguishes DOI-audited experimental evidence from external engineering benchmarks; (6) surrogate-assisted generative design exploration for scenarios where limited literature data cannot resolve multi-objective trade-offs.

4.1 Experimental setup

Dataset and corpus. The knowledge graph is constructed from 75,979 peer-reviewed optoelectronic papers and stores DOI-linked Study and System records covering synthesis routes, fabrication processes, characterization operations, measurement data, and device performance specifications (Section 3.1). Entity extraction performance is evaluated on 11 held-out manuscripts annotated with fact-level labels following the proposed domain ontology. Retrieval experiments adopt 2000 paraphrased requirement queries generated from standardized system-profile cards. An additional system-profile oracle setting is deployed to examine the model's capacity to recover semantically nearest-neighbor material systems.

Baselines. For all ablation variants, the evidence-constrained auditing module is disabled, and a unified Top-K retrieval budget is maintained for fair comparison (Table 1). Four pipeline configurations are defined as follows: B0 generates LLM-paraphrased candidate profiles without graph hydration; B1 implements keyword-driven Neo4j retrieval equipped with graph hydration; B2 performs single-stage MatBERT embedding retrieval on system profiles without keyword-based supplementation; B3 corresponds to our complete cascaded multi-stage pipeline. This set of ablations enables separate comparisons between keyword-based graph retrieval, single-stage semantic retrieval, and their cascaded integration under identical experimental conditions.

Table 1. Parameter configurations of four ablation groups (B0–B3). The audit workflow is disabled for all ablation tests.

Config	Semantic index	Neo4j keyword	Graph hydration	MMR reranking (λ=0.72)
B0	LLM paraphrased profiles	No	No	No
B1	No	Yes	Yes	No
B2	Raw query embedding	No	No	No
B3 (full	System profile	Yes	Yes	Yes

Config	Semantic index	Neo4j keyword	Graph hydration	MMR reranking (λ=0.72)
pipeline)	embedding			

Metrics. For entity extraction, we adopt micro-averaged Precision, Recall and F1 calculated at the fact level based on true positives (TP), false positives (FP), and false negatives (FN). Retrieval performance is quantified via Recall@K and mean reciprocal rank (MRR) against ground-truth (DOI, system ID) pairs, together with metric coverage (MCov). MCov is a binary query-level indicator, assigned as 1 if any retrieved evidence entry contains at least one requested target-metric field within structured records, and 0 otherwise. Notably, MCov only validates the existence of relevant metric fields rather than compliance with numerical performance thresholds; threshold verification and performance gap labeling are executed in the subsequent evidence-constrained auditing stage.

Table 2. Definition of six metric gap status labels for audit workflow and the eligibility for metric leader ranking.

Status	Definition	Eligible as metric leader
Meets target	Value satisfies threshold or matches within tolerance	Yes
Below or Above target	Target threshold unfulfilled	Yes
Off-target	Equality out of tolerance range	Yes
Mentioned only	Metric text exists without numeric data	No
Not reported	No relevant metric records found	No

To comprehensively characterize qualitative differences in Top-K candidate shortlists beyond conventional retrieval metrics, we introduce three supplementary evaluation metrics: UniqueDOI@K, UniqueMaterial@K and intra-list distance (ILD@K). UniqueDOI@K and UniqueMaterial@K denote the counts of distinct DOIs and normalized material strings within the Top-K retrieved entries. As an inherent constraint, UniqueDOI@K ≤ K, while UniqueMaterial@K may exceed K when a single evidence entry involves multiple materials. ILD@K is defined as the average pairwise cosine distance between MatBERT embeddings of Top-K system profiles, which reflects the diversity of retrieved candidate sets.

4.2 Entity extraction quality

Domain researchers defined the ontology schema and manually reviewed extracted Study and System records throughout corpus construction. For quantitative evaluation, we adopt Qwen-Plus as an automated evaluator, which aligns each extracted JSON record with the original manuscripts and labels each fact as TP, FP or FN following the

predefined ontology. Eleven optoelectronic manuscripts are processed via the dual-verification pipeline, with DeepSeek-V4-Pro serving as the default extraction model. Aggregated evaluation yields micro-averaged Precision = 0.990, Recall = 0.952 and F1 = 0.971 (729 TP, 7 FP, 37 FN). The Study and Characterization entities achieve an F1 score of 1.000; Property, Performance and Structure entities obtain F1 values exceeding 0.988. By contrast, Process (F1 = 0.928) and Measurement (F1 = 0.943) exhibit relatively lower recall, which is mainly attributed to fragmented textual descriptions of fabrication procedures and compact tabular measurement data in raw literature.

To validate the reliability of automated evaluation results, a domain expert independently examined a random subset containing 100 extracted facts from the 11 held-out manuscripts against the original full text, following a standardized checklist: entity existence, numerical values consistent with corresponding metrics within a 5% tolerance, and complete multi-token descriptive evidence. Under this rigorous manual protocol, 79 out of 100 extracted facts (79%) are fully supported by source literature. The unsupported entries mostly arise from absent entity information in the raw text corpus or isolated numerical values without associated metric descriptors, instead of arbitrarily generated numerical data.

We emphasize that the manual audit adopts stricter evidence criteria than the Qwen-Plus automated evaluator. The human reviewer demands complete source traceability for every extracted field, while Qwen-Plus only assesses whether extracted information conforms to the structural constraints of the ontology schema. Consequently, the 79% manual support ratio reflects factual completeness and traceability, and does not conflict with the 0.990 extraction precision. The F1 = 0.971 effectively quantifies the extraction performance under ontology constraints, whereas the 21% unsupported fraction illustrates inherent dataset limitations, which are further discussed in Section 6, including data sparsity and incomplete text coverage in scientific publications.

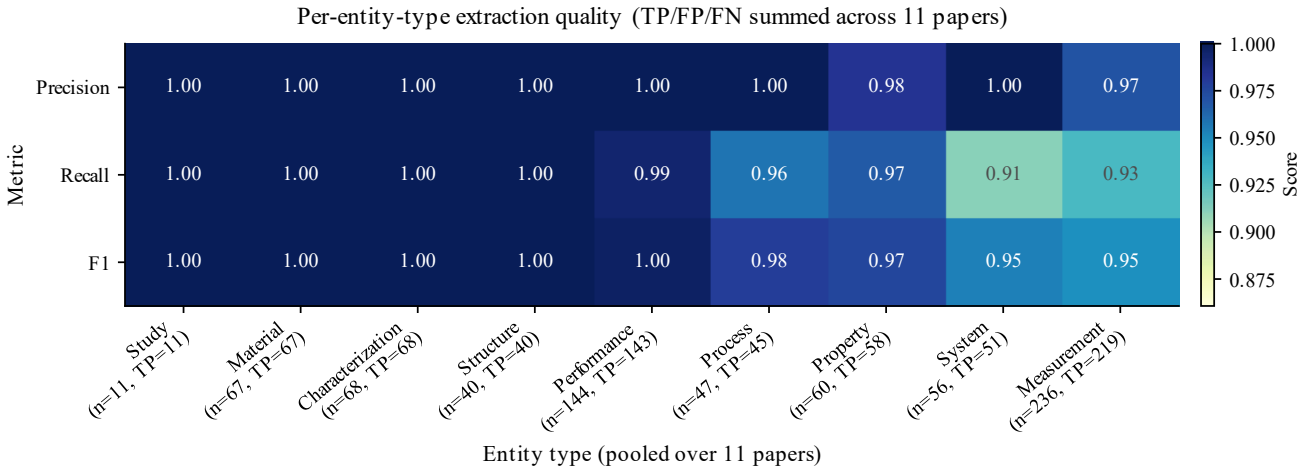


Figure 6. Heatmap of Precision, Recall and F1 scores for each ontology entity category aggregated over 11 annotated manuscripts. The sample count (n) and true positive (TP) count are labeled for each entity type.

4.3 Knowledge-graph statistics and encoder choice

Table 3 presents the detailed node and relation statistics of the constructed OptoMatKG, containing 2.625 million entities and 3.384 million typed relational edges. The designed ontology maintains a standardized

hierarchical chain of *Study–System–Material/Process/Performance*, where each Study node is anchored with original DOI provenance information. This DOI-trackable architecture supports fine-grained attribute retrieval and evidence tracing, enabling reliable factual auditing of characterization and measurement records.

Table 3. Statistics of the OptoMatKG knowledge graph.

Node Type	Count	Relation Type	Count
Property	810,143	has property	810,143
Performance	347,718	test by	852,976
Process	214,990	has performance	347,718
Structure	238,765	determines	44,475
Characterization	135,164	consists of	431,620
Material	431,620	has process	214,990
System	210,619	acts on	71,192
Measurement	160,261	has structure	238,765
Study	76,079	obtained with	161,369
		has system	210,619
Total	**2,625,359**	**Total**	**3,383,867**

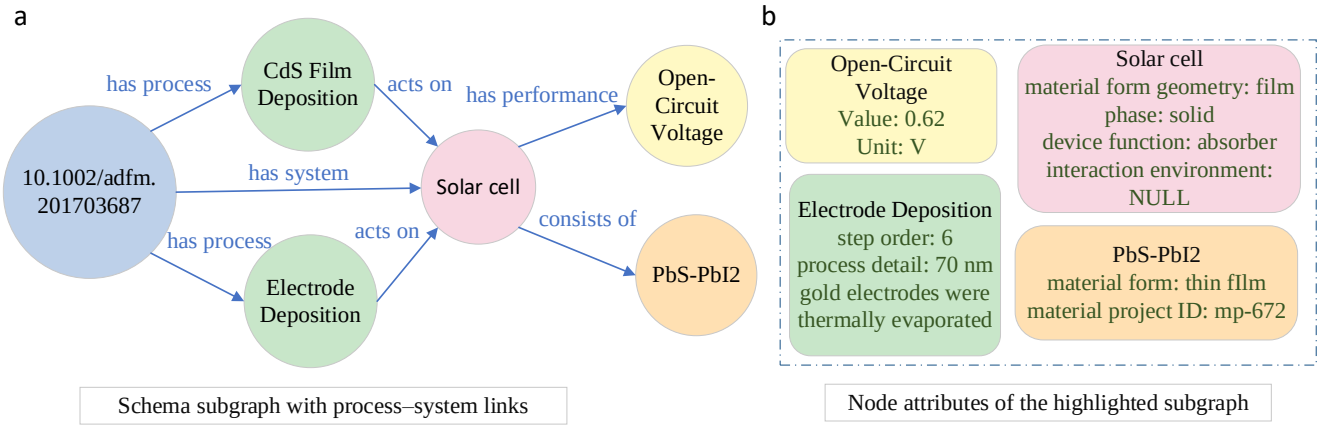


Figure 7. Example OptoMatKG subgraph for a solar cell. (a) Schema links among Study, Processes, System, Material and Performance. (b) Corresponding node attributes, including performance values, system descriptors, process details and Materials Project IDs.

We evaluate the performance of OptoMatKG from five dimensions: entity extraction quality on 11 annotated manuscripts, large-scale knowledge graph statistical characteristics, retrieval ablation experiments on 2000 benchmark queries, fine-grained SiC entity modeling, and a representative photodetector retrieval case study.

All test labels are manually annotated by domain researchers, and we employ Qwen-Plus to conduct automated evaluation: the model compares each extracted JSON record against source manuscripts and labels true positives

(TP), false positives (FP) and false negatives (FN) at the fact level under our ten-category ontology. Eleven optoelectronic manuscripts are processed through the dual-investigator pipeline with DeepSeek-V4-Pro as the default extractor. Aggregated statistics across all samples yield micro-averaged Precision = 0.990, Recall = 0.952 and F1 = 0.971 (729 TP, 7 FP, 37 FN). Study and Characterization entities achieve perfect F1 = 1.000; Property, Performance and Structure exceed F1 = 0.988. Process (F1 = 0.928) and Measurement (F1 = 0.943) exhibit relatively lower recall, attributed to fragmented process descriptions and dense tabular test records in raw text. These results verify that the proposed ontology and extraction pipeline capture most literature-grounded facts required for downstream inverse design tasks.

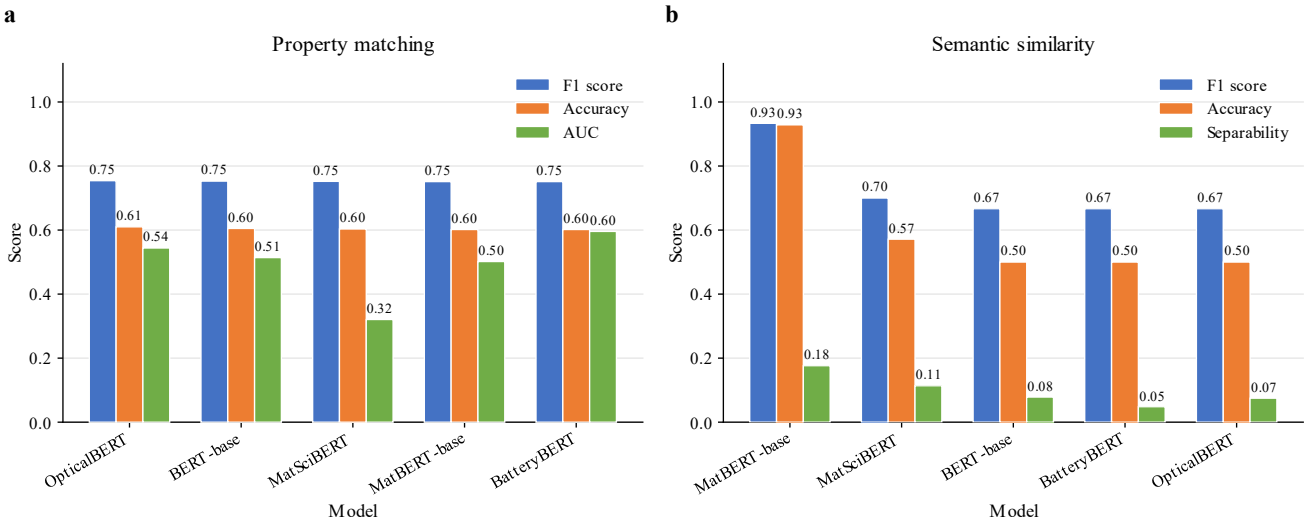


Figure 8. Quantitative comparison of five BERT-based pre-trained models on property matching (a) and semantic similarity (b). F1, accuracy and separability are reported for each model.

To verify OptoMatKG's fine-grained modeling and multi-dimensional association capability, we use a film-type silicon carbide (SiC) case study. The SiC node SiC|mp-8062 is linked via directed relations to five key thin-film structural and morphological properties (grain size, film width, surface roughness, interplanar spacing, and Raman peak position). Each property node stores standardized values, units, characterization methods, an literature sources, enabling systematic aggregation across material–microstructure–characterization evidence. Details are in Table 4.

Table 4. Standardized Attributes of 6 SiC Material Entities in the OptoMatKG Knowledge Graph

No.	Material Name	Material Form	Material Id	mpId
1	SiC	N/A	SiC|mp-11714	mp-11714
2	SiC	Film	SiC|mp-8062	mp-8062
3	SiC	bulk	SiC|	N/A

No.	Material Name	Material Form	Material Id	mpId
4	SiC	thin Film	SiC|mp-11713	mp-11713
5	SiC	N/A	SiC|mp-1002206	mp-1002206
6	SiC	bulk	SiC|mp-1204356	mp-1204356

4.4 Retrieval ablation

We conduct retrieval ablation experiments on 2000 paraphrased optoelectronic design queries to quantify the effectiveness of different pipeline components. Under the unified evaluation protocol, the keyword-enhanced graph retrieval (B1) and the full cascaded pipeline (B3) achieve competitive gold-DOI retrieval performance, with identical core metrics: Recall@10 = 0.486, Recall@50 = 0.614, MRR = 0.303, and a near-complete metric coverage (MCov = 0.998). In contrast, the single-stage MatBERT semantic retrieval (B2) suffers severe degradation in gold-DOI localization, obtaining extremely low ranking performance (Recall@10 = 0.0015, Recall@50 = 0.004, MRR = 0.001) and limited metric coverage (MCov = 0.396). The LLM-only baseline (B0) yields marginal recall performance near the baseline floor (Recall@10 ≈ 0, MCov ≈ 0.33). These results demonstrate that pure semantic embedding matching or LLM-based paraphrasing cannot support reliable DOI-level system retrieval for open-ended design queries, and keyword-driven graph supplementation is indispensable for valid literature grounding in practical material screening scenarios.

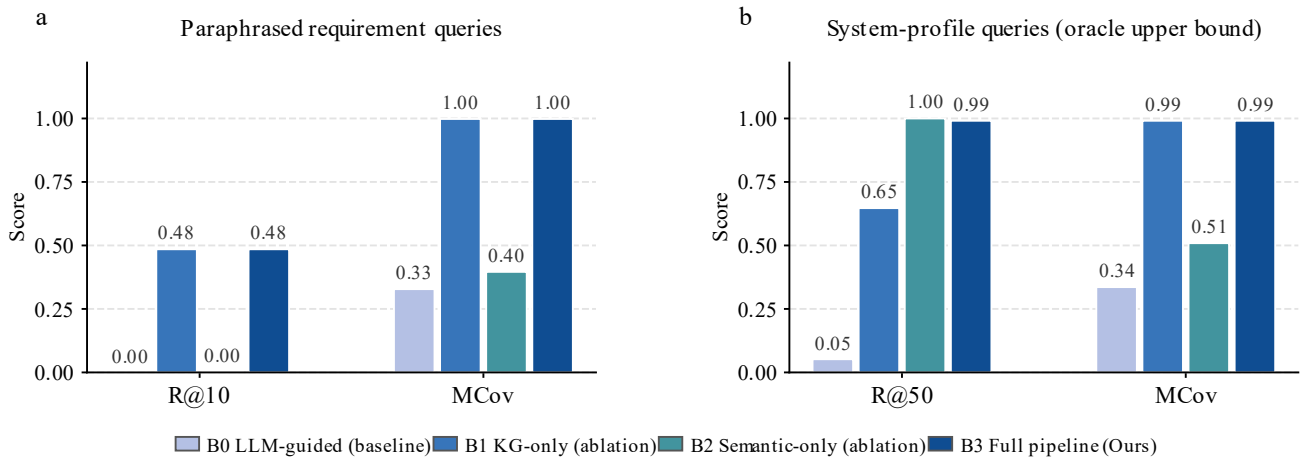


Figure 9. Retrieval ablation results on paraphrased requirement queries and system-profile oracle queries (b). All bars illustrate Recall@K and metric coverage (MCov). MCov evaluates the existence of target metric fields in structured evidence rather than numerical threshold compliance.

Despite the comparable gold-recall performance between B1 and B3 on paraphrased requirement queries, the two pipelines generate fundamentally different candidate shortlist compositions. The consistent Recall@K, MRR and MCov results indicate that literal keyword matching dominates exact gold-DOI retrieval under the single-gold evaluation protocol adopted in this work. Rather than pursuing marginal improvements in exact gold recall, the cascaded multi-stage design of B3 focuses on optimizing the quality and diversity of retrieved candidate pools. Statistical results on all 2000 query samples show that B3 produces a significantly larger volume of valid evidence entries (average ≈50 entries per query) than the pure keyword-based B1 pipeline (average ≈25 entries), at the cost of moderate computational latency. Specifically, the integration of semantic indexing and MMR diversification rebalances the ranking distribution: B3 weakens the over-concentration of keyword-matched systems in the Top-10 ranking while maintaining exact gold-DOI recall accuracy. Since the OptoMatKG framework incorporates a downstream evidence-auditing module instead of directly adopting raw Top-K results, the expanded and diversified candidate pool generated by B3 provides richer structured evidence for subsequent gap analysis and factual auditing.

Further diversity verification results validate the inherent advantages of the full pipeline (Figure 10). At K=10, B1 and B3 exhibit comparable performance in UniqueDOI@K, UniqueMaterial@K and intra-list distance (ILD@K). When expanding to K=50, B3 substantially increases the number of distinct DOIs and normalized material candidates while maintaining competitive embedding diversity, verifying that the cascaded semantic-keyword fusion mechanism effectively breaks the limitation of keyword-only retrieval and enriches candidate coverage. In contrast to paraphrased requirement queries, system-profile oracle queries (Figure 9b) with standardized and profile-consistent wording fully demonstrate the superiority of the full pipeline. B3 achieves a much higher Recall@50 of 0.991 compared with 0.647 for B1, while preserving a high metric coverage of 0.991. Overall, exact gold-DOI retrieval for open-ended user queries is mainly constrained by keyword matching, while the proposed multi-stage pipeline delivers core values in expanding diverse audit candidates and improving semantic matching accuracy for standardized profile queries.

We further analyze the practical significance of the obtained Recall@10 = 0.486 result. This metric indicates that nearly half of the paraphrased queries fail to retrieve the exact gold-standard DOI within the Top-10 candidates under the single-gold evaluation setting. We treat this result as an inherent limitation of the benchmark protocol rather than a defect of the system. Different from conventional closed-domain question-and-answer models that pursue exact single-answer matching, OptoMatKG is designed as an evidence-auditable material screening system. Its core practical value lies in providing traceable DOI-level evidence, complete metric field coverage (MCov = 0.998), reliable long-tail retrieval capability (Recall@50 = 0.614), and explicit unreported-metric gap labeling. Even if the exact gold record is missing from the Top-10 ranking, the retrieved candidates still provide valid, verifiable structured evidence for manual inspection, screening and iteration, which conforms to the standard workflow of scientific material design.

In-depth error analysis reveals three main causes for Top-10 false-negative cases: first, semantic gaps exist between free-form user query paraphrases and the standardized structured fields of indexed system profiles; second, user queries often contain underspecified or implicit constraint information, while the gold-standard system profile records contain richer device and process attribute details; third, the strict single-(DOI, system) labeling protocol

treats highly similar and homologous literature as invalid matches, even when these candidates contain valid auditing metrics consistent with user requirements. We randomly sampled 50 B3 false-negative cases for quantitative statistics: 16% of errors stem from query paraphrasing ambiguity and incomplete constraint description; 26% correspond to near-miss cases where the gold-standard system is ranked between 11–50; 58% belong to deep-miss cases where the target system exists in the knowledge graph but fails to enter the Top-50 ranking. Future improvements will focus on enhancing LLM-based query normalization and entity grounding, and relaxing the single-gold evaluation constraint to adapt to multi-evidence scientific retrieval scenarios.

In oracle query scenarios with highly consistent textual distribution with indexed system profiles, single-stage semantic retrieval (B2) achieves near-perfect Recall@50 but suffers a sharp drop in metric coverage (MCov = 0.509). By comparison, B1 and B3 maintain superior metric field coverage (MCov = 0.991) with competitive retrieval accuracy. The comprehensive performance comparison demonstrates that the proposed cascaded pipeline realizes a practical and robust trade-off: semantic embedding ensures fine-grained semantic discovery, keyword-graph fusion guarantees structured metric field alignment, and MMR diversification optimizes candidate diversity for downstream evidence auditing.

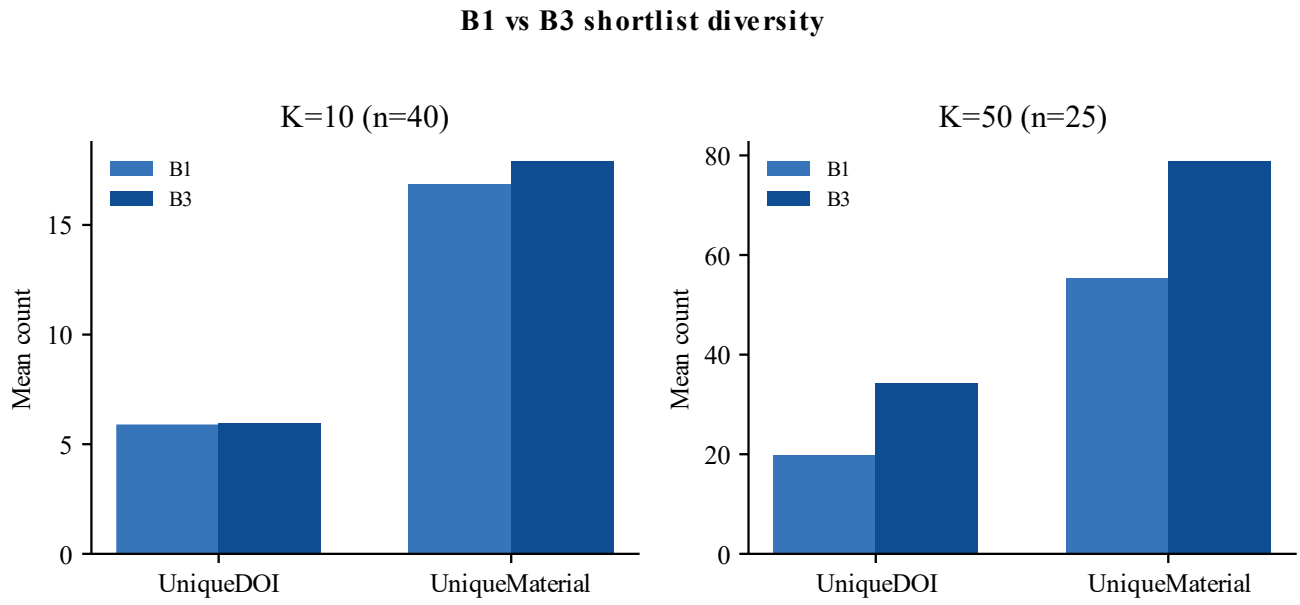


Figure 10. Candidate shortlist diversity comparison between B1 and B3. The statistical indicators include average UniqueDOI@K and UniqueMaterial@K . The two pipelines show comparable Top-10 diversity, while B3 achieves significantly expanded unique DOI and material coverage at K=50 (n = 40 for K=10; n = 25 for K=50).

4.5 Fine-grained modeling and photodetector case study

We evaluate the system against a representative design requirement: "photodetector operating at 300 K and 1550 nm, with specific detectivity $D^* > 5 \times 10^{11}$ Jones, response time < 50 ps, and dark current < 1 nA." OptoMatKG retrieves candidate systems via knowledge graph traversal, performs deep attribute resolution, and assesses metric satisfaction against DOI-linked quantitative records. Among the retrieved literature entries, the highest reported

specific detectivity is $D^* = 3.5 \times 10^{10}$ Jones, observed in a GaInAs/InP quantum-well infrared photodetector—falling short of the target threshold. Moreover, response time and dark current values are frequently unreported in the literature, and wavelength-based retrieval can return off-target candidates (e.g., ultraviolet AlGaN photoconductors). Rather than synthesizing fictitious devices that fully meet all constraints, the auditing module returns partially matched, evidence-grounded candidates and explicitly flags missing performance metrics.

External engineering benchmarks and surrogate design hypotheses. Distinct from DOI-grounded evidence evaluation, we introduce external references for comparative context. We adopt published performance ranges from commercial datasheets for lattice-matched $In_{0.53}Ga_{0.47}As$ as practical benchmarks: responsivity ≈0.85–0.95 A W^{-1}, dark current spanning pA to nA, response time ≈35 ps, and $D^* > 1 \times 10^{12}$ Jones at 1550 nm under 300 K. Separately, we examine a waveguide PIN device concept based on $Ge_{0.829}Sn_{0.171}$ through surrogate device modeling. This composition offers a bandgap of ~0.774 eV, closely matching the 1550 nm target, and yields a projected response time of ~5.4 ps in a narrow-depletion design. However, the lattice mismatch with Si introduces inherent penalties, including elevated dark current and reduced detectivity. We stress that these surrogate projections and commercial reference data are neither embedded into OptoMatKG's structured metric fields nor claimed as experimentally validated, DOI-audited findings.

4.6 Generative design exploration via surrogate-assisted GA

When DOI-audited literature screening yields only partial evidence and cannot resolve inherent multi-objective performance trade-offs, OptoMatKG activates a surrogate-assisted NSGA-II evolutionary optimization pipeline (Section 3.4) for hypothesis-driven design exploration. This generative module explores feasible device configurations at the proxy level rather than fabricating non-existent literature records. The optimization evolves normalized design vectors spanned by key GeSn waveguide PIN parameters, including material composition, strain magnitude, depletion width, waveguide length, and active area. After decoding into physical units, candidate designs are evaluated via a multi-objective surrogate function, without altering the DOI-retrieved evidence shortlists or optimizing experimentally reported literature values. As a complementary module, the generative exploration workflow augments, rather than replaces, the evidence-constrained screening paradigm.

The surrogate multi-objective loss vector is defined as:

$$f(x) = \left[\Delta E_g(x), \tau_{ex}(x), I_{dark,ex}(x), D^*_{sf}(x)\right].$$

Device performance metrics are calculated through analytical surrogate models tailored for GeSn waveguide PIN architectures, including optical absorption $\eta = 1 - \exp(-\alpha L)$, combined transit/RC response time $\tau = \sqrt{\tau_{\mathrm{tr}}^2 + \tau_{\mathrm{RC}}^2}$, area- and perimeter-coupled dark current $I_{\mathrm{dark}} = J_0 A + c\sqrt{A} J_0$, and specific detectivity $D^* = R\sqrt{A}/\sqrt{2qI_{\mathrm{dark}}}$. Full formula derivations and calibrated coefficients are provided in the Supporting Information. While the surrogate formulas and objective calibrations are customized for the GeSn photodetector case, the overall NSGA-II optimization framework is fully reusable and can be rapidly adapted for other material and device systems via surrogate replacement.

The bandgap mismatch term is derived from a normalized fitting score targeting the 1550 nm corresponding bandgap $E_{g,\text{target}} \approx 0.800$ eV, with a tolerance range $\delta = 0.25$ eV:

$$fit = \max\left(0, 1 - \frac{|E_g(x) - E_{g,target}|}{\delta}\right), \Delta E_g(x) = 1 - \text{fit}.$$

The remaining three objectives adopt one-sided, target-normalized excess/shortfall formulations clamped within the range $[0, 1]$ to ensure balanced multi-objective optimization:

$$\tau_{\text{ex}} = \min(1, \max(0, \frac{\tau - \tau_t}{\tau_t})),$$

$$I_{\text{dark,ex}} = \min(1, \max(0, \frac{I_{dark} - I_t}{I_t})),$$

$$D^*_{\text{sf}} = min(1, max(0, 1 - \frac{D^*}{D^*_t})).$$

Notably, to avoid objective saturation and preserve meaningful Pareto trade-offs, the GA optimization does not adopt the original commercial-level query thresholds (50 ps, 1 nA, 5×10^{11} Jones). Instead, the objective targets are recalibrated to the effective operating range of the device surrogate model: $\tau_t = 4$ ps, $I_t = 50$ nA, and $D^*_t = 3.4 \times 10^7$ Jones.

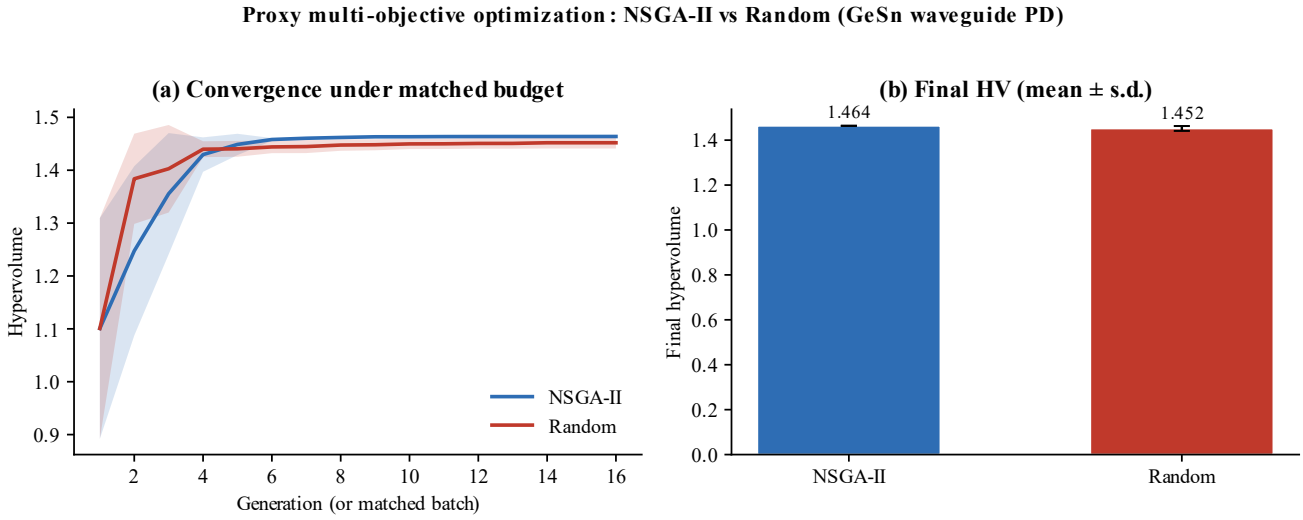


Figure 11. Multi-objective optimization benchmark between NSGA-II and random sampling based on the GeSn waveguide photodetector surrogate model.

We evaluate the exploratory performance of NSGA-II under fixed and matched computational budgets: population size of 20, 15 evolutionary generations, and approximately 320 function evaluations per random seed, with five independent repeated runs. The proposed NSGA-II pipeline achieves a higher mean hypervolume of the Pareto archive than random sampling (1.4639 ± 0.0001 vs. 1.4519 ± 0.0109) and yields lower scalarized surrogate loss (Figure 11). Consistent performance improvement is observed across all five seed runs. A paired two-sided Wilcoxon signed-rank test yields $p = 0.0625$, which represents the theoretical minimum achievable two-

sided p-value for five identically signed paired differences. A paired t-test further returns $p = 0.069$. Neither result satisfies the standard statistical significance threshold ($\alpha = 0.05$). Therefore, the obtained results demonstrate a stable and promising exploratory trend under limited computational budgets, rather than statistically definitive superiority. This observation aligns with the module's design purpose of surrogate-assisted generative exploration. All optimized solutions are purely model-derived and do not constitute physical experimental validation.

5 Conclusion

This work develops OptoMatKG, a literature-grounded knowledge graph integrated with a cascaded retrieval-and-audit framework for evidence-constrained optoelectronic material screening. Constructed from 75,979 peer-reviewed publications, the established knowledge graph contains 2.625 million entities and 3.384 million typed relational edges, forming a structured, DOI-traceable data foundation for device-centric knowledge organization. Integrating MatBERT semantic indexing and multi-stage graph fusion, the framework converts open-ended natural-language design queries into standardized, auditable evidence packages. These structured outputs deliver ranked candidate systems, quantitative performance gap annotations, fine-grained graph attributes, and interpretable material-process-performance reasoning chains grounded in literature. Quantitative evaluation demonstrates that the entity extraction pipeline achieves a micro-averaged F1 score of 0.971 on manually annotated manuscripts. Evaluated on 2000 practical design queries, the keyword-graph-enhanced pipeline significantly outperforms single-stage semantic embedding retrieval in exact DOI-level system grounding, while the high metric coverage guarantees field-aligned structured evidence for downstream auditing, avoiding unconstrained free-form generation.

Comprehensive case studies further validate the framework's practicality and robustness. The thin-film SiC subgraph analysis verifies the model's capacity for fine-grained, multi-dimensional property association and complete literature provenance recording. The 1550 nm photodetector screening case demonstrates that OptoMatKG reliably outputs partially matched literature evidence and explicitly identifies unresolvable performance gaps, without generating fictitious fully compliant device records. Furthermore, surrogate-assisted genetic algorithm exploration and commercial datasheet benchmarks are strictly categorized as model-derived engineering hypotheses, which are clearly decoupled from literature-verified experimental evidence to maintain factual rigor.

We systematically analyze the limitations of the present system. Under the strict single-gold-DOI evaluation protocol, the best-performing cascaded pipeline (B3) achieves comparable Top-10 recall performance to pure keyword graph retrieval (B1), indicating that exact gold-DOI localization remains dominated by literal keyword matching for paraphrased user queries. Accordingly, practical screening effectiveness should be comprehensively evaluated by integrating metric coverage, candidate pool diversity, oracle-query retrieval accuracy, deep Recall@K performance, and auditable gap labeling, rather than relying solely on top-ranked exact matching. A domain expert checklist-based audit on 100 randomly sampled extracted facts validates that 79% of the structured records are fully traceable to original full-text content. In-depth error analysis further attributes most Top-10 false-negative cases to deep ranking failures and near-miss gold rankings, rather than incomplete profile coverage in the knowledge base.

Despite the inherent coverage limitation of the finite literature corpus, OptoMatKG exhibits distinct advantages over conventional retrieval-augmented generation (RAG) systems. It strictly constrains all numerical performance reports to DOI-verified graph evidence, explicitly marks unreported metric fields, and systematically separates model-generated engineering hypotheses from literature-supported factual claims. Different from traditional open-ended generation paradigms, this framework prioritizes traceability, auditability and factual reliability for scientific material screening. Future directions include enhancing query normalization and paraphrase adaptation, improving top-rank candidate diversity, expanding evolutionary optimization budgets to achieve statistically significant generative exploration, strengthening cross-study relational reasoning, and releasing a public query API to facilitate open-source data science applications.

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Author Contributions The author has given approval to the final version of the manuscript.

Data Availability Statement Code and data are available at: https://github.com/ni-zefeng/OptoMatKG

Declarations

Conflict of Interest

The authors have no conflicts to disclose.

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